Individual Addendum: Yuri Rodriguez **Name:** Yuri Rodriguez **Team:** ILOVECODING **Date:** May 1, 2026 --- ## 1. Personal Contribution My contributions to the ILOVECODING capstone project focused on data integration across sources, exploratory trend analysis, diagnostic testing implementation, and final communication of findings for portfolio management audience across all four milestones. - **Milestone 1 (FRED Data Integration & Time-Series Alignment):** Retrieved Federal Reserve Economic Data (Federal Funds Effective Rate, Treasury yields) and merged with REIT panel using month-year keys; verified temporal alignment (no missing months, correct date indexing); created macro-economic control variables (Fed rate changes, yield curve slope); documented FRED series IDs and update frequency. **14 hours** - **Milestone 2 (Trend Analysis & Sector Patterns):** Analyzed REIT sector returns over time (2000–2024) with event-marking (financial crisis, QE periods, rate hikes); created time-series plots showing Federal Funds vs. REIT returns correlation; identified sectoral divergence (Industrial vs. Retail performance gaps); drafted preliminary findings on macro-financial relationships. **13 hours** - **Milestone 3 (Diagnostic Implementation & Model Testing):** Implemented full diagnostic suite: Breusch-Pagan heteroskedasticity test, VIF collinearity checks, Durbin-Watson autocorrelation test, Q-Q normality assessment; ran specification tests comparing alternative lag structures; generated diagnostic plots for residual analysis. **14 hours** - **Milestone 4 (Investment Memo Communication & Final Presentation):** Co-authored Investment Recommendations section with practical portfolio application (beta tilting by rate scenario); wrote clear economic interpretations (61 bps = 7.3% annualized for business audience); created honest caveats section emphasizing data limitations; prepared final memo for PDF submission. **14 hours** **Total Capstone Hours:** ~55 hours --- ## 2. One Defended Methodological Decision *Decision: Use Federal Funds Effective Rate rather than market expectations (fed funds futures) as the primary interest-rate control variable.* **Reasoning:** In Milestone 1, we needed to capture how interest-rate movements affect REIT returns. Two choices emerged: (1) actual Fed Funds rate realized ex-post, or (2) market expectations embedded in fed funds futures prices. Our choice of actual rates reflects the following logic: - **Realized rates are exogenous to daily REIT trading.** Monthly REIT returns are aggregates across 22 trading days. The Federal Reserve sets the Federal Funds target at discrete dates (typically 8 FOMC meeting days per year); between meetings, rates drift based on open-market operations. Using realized ex-post rates avoids reverse causality: large REIT portfolio flows cannot influence the Fed's announced target. - **Data availability and reliability.** Fed Funds Effective Rate (FRED series FEDFUNDS) has 99+ years of daily data with no missing values. Fed futures data, while available, requires careful splicing across contract expirations and introduces sampling complexity. For a 2000–2024 panel with 34,121 observations, the simplicity and reliability of realized rates were paramount. - **Economic interpretation clarity.** When we say "higher realized Fed rates

correlate with lower REIT returns," the mechanism is clear: after the Fed tightens (increasing borrowing costs), REITs refinance at higher rates, reducing economic value (discount rate effect). Using expectations would conflate expectations revisions (which may be surprising news) with actual rate increases. Evidence from M2 corroborates this choice: realized Federal Funds rate shows $r \approx -0.38$ with REIT returns (inverse relationship), consistent with economic theory. This choice is robust and interpretable. --- ## 3. One Key Limitation *Limitation: Macroeconomic variables (Federal Funds Rate, Treasury yields) are time-invariant within each month, creating potential confounding with time fixed effects.* **Why It Matters:** Our model includes time fixed effects (δ_t) that absorb all month-specific shocks. The Federal Funds Rate is constant across all 273 REITs within a month—it's a macro variable. If we were to add the Fed rate as a standalone regressor alongside time fixed effects, it would be perfectly collinear (multicollinearity: months with identical Fed rates would produce singular design matrix). This means our model cannot separately identify leverage-return effects from interest-rate level effects; time fixed effects absorb the macro effect. **Impact on Findings:** - **Beta coefficient preserved.** Beta (systematic risk) is firm-specific and varies across REITs within each month. It remains identified even with time fixed effects. Our t=4.46 on beta is robust. - **Interest-rate mechanism obscured.** We cannot directly estimate "a 100 bps increase in Fed rates reduces REIT returns by X%." The mechanism operates through two channels: (1) discount rate (time FE absorbs macro component), (2) leverage channel (our leverage variable captures this). By using lagged leverage, we proxy for interest-rate pass-through. - **Causal interpretation limited.** Our coefficient on leverage reflects "change in leverage predicts returns" but conflates leverage itself with interest-rate expectations (if REITs increase leverage expecting lower rates, that's forward-looking behavior, not causal effect of leverage on returns). **How to Address:** An alternative approach—use cross-sectional variation in interest rate exposure (some REITs face fixed rates, others variable; varies by firm)—would isolate true leverage effects separately from interest-rate cycles. This requires more granular debt structure data (fixed vs. floating rate breakdown by REIT), which was unavailable for our analysis. Our limitation is data-constrained rather than methodological. --- ## 4. AI Audit Notes **Task 1: Diagnostic test implementation (Breusch-Pagan, Durbin-Watson, VIF)** - **Prompt:** "Provide Python code to run heteroskedasticity test, autocorrelation test, and multicollinearity checks on panel regression output." - **Output:** AI provided statsmodels code for all three diagnostics with correct syntax. - **Verification:** Ran on actual M3 regression results. Breusch-Pagan $\chi^2 = 14.2$ ($p=0.003$), Durbin-Watson ≈ 1.95 (no autocorrelation), VIF max ≈ 2.1 (no collinearity). All results manually spot-checked against statistical software outputs and hand calculations for subset of variables. - **Changes:** Added interpretation notes. AI provided p-values but didn't contextualize: clarified that Breusch-Pagan $p<0.05$ means "heteroskedasticity detected, mitigated by entity clustering" and DW ≈ 2 means "no serial correlation in residuals." **Task 2: Time-series visualization (Federal Funds vs. REIT returns)** - **Prompt:** "Create dual-axis plot showing Federal Funds Effective Rate (left axis) and average REIT monthly returns (right axis) over 2000–2024." - **Output:** AI produced matplotlib code with proper dual-axis scaling. - **Verification:** Plot created successfully. Visual inspection shows inverse relationship, especially during tightening cycles (2004–2008, 2022–2023). Calculated correlation coefficient (r = −0.38) to quantify relationship and verified alignment with visual pattern. AI didn't provide correlation; I added it. - **Changes:** Enhanced plot with event annotations (2008 crisis, beginning of rate hiking cycles) to clarify interpretation of visual patterns. **Task 3: Portfolio recommendation prose** - **Prompt:** "Write investment recommendations based on econometric findings that translate regression coefficients into actionable portfolio tilts for a professional investor." - **Output:** AI generated template recommendations (overweight/underweight sectors, beta tilting) with generic language. - **Verification:** Recommendations were structurally sound but lacked specificity. Cross-referenced against M2 and M3 findings (Industrial sector avg return 1.14%, Retail 0.91%) to ground recommendations in empirical results. Verified sector allocations sum to 100%. - **Changes:** Added specific percentage tilts (Overweight Industrial +15%, Underweight Retail −15%) and economic rationale (leverage stability, rent growth trends) from our sector analysis. **Overall Assessment:** AI provided correct statistical and coding scaffolds but required human interpretation of results and business contextualization. Numerical findings verified against original M1–M3 computational outputs. All statistical tests and visualizations reproduced from raw data or regression output. --- **Status:** ✓ COMPLETE **Submission Date:** May 1, 2026 by 11:59 PM EST